Introduction to IT–ITeS Industry

The big idea

IT (Information Technology) is the use of computers, software and networks to store, process and send information. **ITeS (IT-enabled Services)** are services delivered using IT, such as call centres, data entry, medical transcription and online customer support. India is one of the world's biggest IT–ITeS hubs, with cities like **Bengaluru**, Hyderabad, Pune, Chennai, Noida and Gurugram, and the sector is a major source of jobs and exports.

IT vs ITeS

IT	ITeS
Building and maintaining software, hardware and networks	Delivering business services *using* IT
Software development, app design, cyber security, cloud	BPO/call centres, data entry, medical transcription, back-office accounting

- **BPO (Business Process Outsourcing)**: a company hands routine work (customer support, billing, payroll) to another firm.
- **KPO (Knowledge Process Outsourcing)**: outsourcing of knowledge-based work needing expertise – research, data analytics, legal and financial analysis.

IT applications in daily life

At home: online shopping, bill payments, video calls, streaming, smart devices.

In education: e-learning platforms (DIKSHA, SWAYAM), smart classes, online tests, digital libraries.

In banking: **ATMs**, net banking, mobile banking, **UPI** payments, core banking.

In healthcare: electronic health records, **telemedicine** (consulting a doctor online, e.g. eSanjeevani), scanning machines (CT, MRI), online appointments.

In governance (e-Governance): online services for citizens – **DigiLocker** for documents, online tax filing, passport application, Aadhaar-based services, **UMANG** app.

In transport: online railway and air ticket booking (IRCTC), GPS navigation, cab apps, **FASTag** toll payment.

In retail and business: billing with barcode scanners, inventory management, e-commerce websites.

In entertainment and communication: social media, online games, OTT platforms, e-mail.

ONE TRIP, MANY IT APPLICATIONS

You book a train ticket on IRCTC, pay through UPI, get an SMS with your PNR, track the train on an app and show the e-ticket on your phone. That is IT in action at every step.

New technologies in IT

- **Cloud computing** – storing and using data and software over the Internet.
- **Artificial Intelligence (AI)** – machines that can learn and make decisions (chatbots, voice assistants).
- **Internet of Things (IoT)** – everyday devices connected to the Internet (smart bulbs, smart watches).
- **Big Data** and **data analytics**.
- **Cyber security** – protecting systems from attacks.

Careers in IT–ITeS

Software developer, web designer, data entry operator, customer care executive, network administrator, data analyst, graphic designer, IT support technician, cyber security analyst.

IT IS NOT ONLY PROGRAMMING

Many IT–ITeS jobs need good communication and keyboarding skills rather than coding – e.g. customer care executive or data entry operator.

EXAM TIP

For "applications of IT" questions, write the **field** as a sub-heading and give one specific example (e.g. Banking – ATM, UPI).

Quick recap

- IT = computers + software + networks for information. ITeS = services delivered using IT.
- BPO handles routine processes; KPO handles knowledge-based work.
- IT is used in home, education, banking, health, governance, transport and retail.
- New trends: cloud, AI, IoT, big data, cyber security.

Practice questions

1. ITeS stands for
   (a) Information Technology for Every School (b) Information Technology enabled Services (c) Internet Technology and e-Services (d) Integrated Technology Services
2. Outsourcing of research and data analysis work is called
   (a) BPO (b) KPO (c) ATM (d) GPS
3. Consulting a doctor through a video call is an example of
   (a) e-commerce (b) e-banking (c) telemedicine (d) data entry
4. DigiLocker is an example of IT used in
   (a) entertainment (b) retail (c) transport (d) e-governance
5. Smart bulbs and watches connected to the Internet are part of
   (a) Cloud computing (b) Big Data (c) Telemedicine (d) Internet of Things
6. Which Indian city is widely known as the "Silicon Valley of India"?
   (a) Bengaluru (b) Kolkata (c) Jaipur (d) Bhopal
7. Differentiate between BPO and KPO with one example each.
8. Describe any three applications of IT in daily life.

Answers: 1. b 2. b 3. c 4. d 5. d 6. a 7. see notes 8. see notes